LeakScope: A Research Artifact for Android Lifecycle Violations

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Artifact

LeakScope turns Android lifecycle obligations into **explainable static warnings** and **reusable research evidence**.



created

static-analysis workbench

used

labels and baselines for the dissertation

next

hybrid detector + repair hints

What the artifact detects

13
detectors

3
families

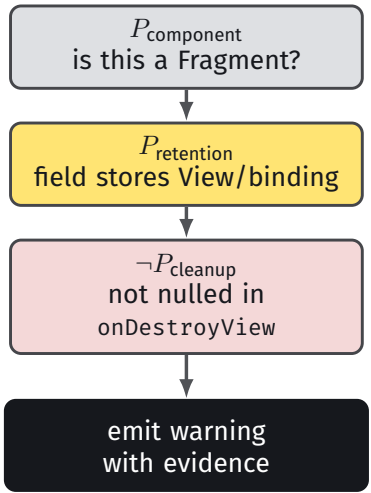
Fragment / View: view field retention, view-binding opportunities, Compose lifecycle misuse

Async ownership: threaded UI references, ViewModel context references, Flow retention, state-holder leaks

Resources / callbacks: BroadcastReceiver work, service resources, callback registration, BillingClient, Loader, Collator

Scope: patterns where Android lifecycle ownership decides whether a reference, callback, or resource should be released.

Inside one rule: a predicate cascade




```
if (!isFragment(cls)) return "";          Pc

if (inViewLifecycle &&
    isViewCreation(rhs))
  viewLocals.add(lhs);

if (isViewField(field) && isViewValue)
  retained.put(field, methodName);        Pr

if (isDestroyView && rhs == null)
  cleared.add(field);                     Pclean
```

What a finding looks like

```
Fragment View Field Retention Leak Detected
Class:  ...fragment.MusicianFragment
Issue:  onDestroyView() is missing
Leaked: view : android.view.View
        assigned in onCreateView

Fix: override onDestroyView() {
       super.onDestroyView();
       view = null;
     }
```

what retained

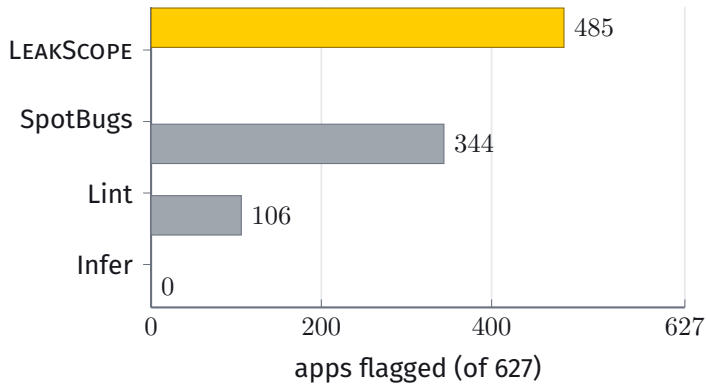
where assigned

why lifecycle mismatch

how to repair

The same evidence is useful to developers, manual validators, and downstream code models.

What the artifact produced so far



734

app–category pairs no other tool reports

7/9

categories with zero overlap

3,152

Yes/No prompts generated for model study

Reader In Mind

Find Real heap leaks with the tool

How to Write a .MD file interactively

How to Integrate it in IDE

LeakScope testing on multiple Platforms.